

End-to-End Voice Bot System Design (Low Latency, Dynamic, Hindi/Hinglish)

1. Problem Statement:

Build a telephonic survey bot that handles dynamic user responses, mixed intents, Hindi/Hinglish input, and maintains latency under 1 second.

2. Core Principles:

- Controlled natural conversation (not open chat)
- Slot filling instead of strict Q&A;
- Intent-aware processing
- Hybrid system (rules + small LLM fallback)

3. Architecture Overview:

Speech → ASR → Intent Detection → Slot Extraction → State Manager → Dialogue Manager → Response Composer → TTS

4. State (Slot) Design:

loan_taken, payment_done, payment_date, payment_mode, amount, payee_name, compliance_flag

5. Intent Types:

- answer
- question (why, who, etc.)
- objection (don't want, busy)
- confusion
- irrelevant

6. Mixed Input Handling:

Example: '2000 diya why asking so many questions'

→ Extract: amount=2000

→ Intent: objection

→ Response: acknowledge data + handle objection + continue flow

7. Response Strategy:

[Acknowledge] + [Handle Intent] + [Next Question]

8. Latency Optimization:

- Avoid large LLM prompts
- Use regex + keyword mapping
- Use micro-LLM only when needed

- Voice pipeline optimized